**chosen, the result is that DBSCAN partitions data perfectly into two concentric rings.
(right) When the radius ε is set to a value so big that the dense area of inner points extends to points of the outer ring, then the whole dataset is mistakenly assigned to the same cluster.**

Hyperparameters[18] tuning, as it often happens in machine learning, is crucial and not always easy. Setting `minPoints` is usually straightforward: as a rule of thumb, for `d`-dimensional datasets, you choose `minPoints > d` and values around `2*d` are often ideal; for particularly noisy datasets it is also recommended to use larger values of this parameter, to strengthen the noise filtering. Conversely, determining the right value for ε is often challenging and requires either deep domain knowledge, or extensive tuning; figure 12.13 shows, by keeping fixed `minPoints`, how values of ε too small (for a given dataset) causes an excessive fragmentation of the dataset into small clusters, while values too large have the opposite effect, reducing the ability of the algorithm to spot different clusters.

Clearly, the need for such a tuning would bring us back to a situation similar to using k-means, where we needed to know in advance how many clusters we wanted; and while in this 2-dimensional example it might seem easy to find what the "right" number of clusters should be, when we move to higher-dimensional spaces, we lose the possibility to use our intuition, and determining the right values of the algorithm hyperparameters based on the number of clusters becomes impossible.